DETAILED DIAGNOSTIC REPORT (DDR)

AI-Generated Property Health Assessment

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DETAILED DIAGNOSTIC REPORT (DDR)

Inspection Date: 27.09.2022, 14:28 IST **Inspected By:** Krushna & Mahesh **Property Type:** Flat (11 Floors) **Previous Structural Audit:** No **Previous Repair Work:** No **Inspection Score:** 85.71% | Flagged Items: 1

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1. PROPERTY ISSUE SUMMARY

The property exhibits widespread dampness and seepage affecting multiple areas including the Hall, Bedroom, Master Bedroom, Kitchen, Parking Area, and Common Bathroom. The inspection reveals a pattern of moisture ingress primarily originating from two sources: internal bathroom plumbing failures and external wall envelope breaches.

Key findings include tile hollowness in bathroom floors, open tile joints allowing water penetration, damaged Nahani traps, loose and corroded plumbing fixtures, and moderate cracking on external wall surfaces. The leakage is reported as continuous ("All time"), indicating active and persistent water ingress rather than seasonal or intermittent issues.

Thermal imaging conducted with a Bosch GTC 400 C Professional device confirmed temperature differentials across affected areas, with hotspot readings ranging from 25.2°C to 28.8°C and coldspot readings from 20.2°C to 23.4°C, consistent with moisture presence within building materials.

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2. AREA-WISE OBSERVATIONS

Impacted Area 1: Hall

- **Negative Side (Damage):** Skirting-level dampness observed along hall walls, indicating prolonged moisture exposure at the base.

Source: Sample Report | Page 1

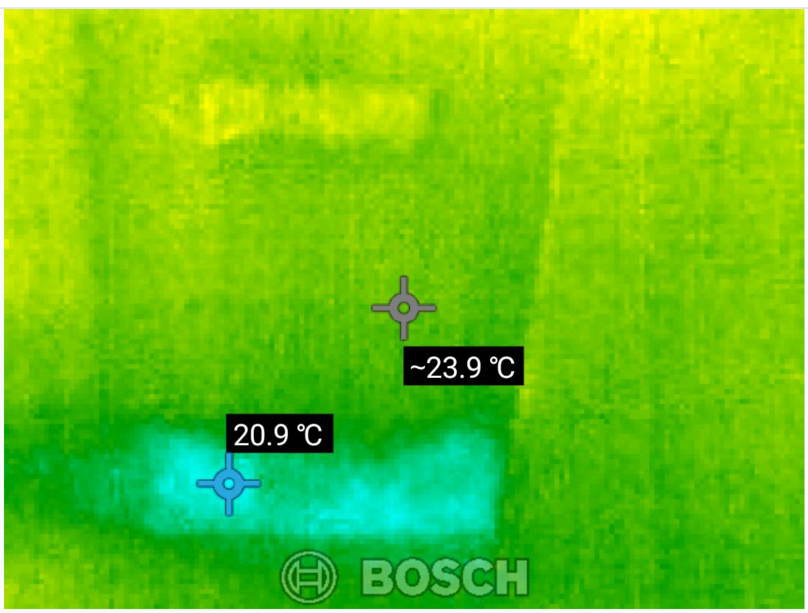
Source: Sample Report | Page 2

- **Positive Side (Source):** Traced to Common Bathroom — tile hollowness detected, allowing water to seep through the floor substrate and travel to adjacent hall walls.

Source: Sample Report | Page 3

Source: Sample Report | Page 3

- **Thermal Findings:** Thermal scans in this zone show temperature range of 22.0°C to 27.0°C, with cooler zones indicating moisture-saturated areas.



Source: Thermal Images | Page 1

Impacted Area 2: Bedroom

- **Negative Side (Damage):** Skirting-level dampness observed, similar pattern to the Hall. No dedicated negative-side photographs were provided in the inspection report for this specific area.

- **Positive Side (Source):** Same source as Hall — Common Bathroom tile hollowness. The bedroom shares a wall with the bathroom, and water is migrating through the masonry.

Source: Sample Report | Page 3

Source: Sample Report | Page 3

- **Thermal Findings:** Temperature differential of approximately 5°C observed between hot and cold spots, consistent with subsurface moisture.

Source: Thermal Images | Page 1

Impacted Area 3: Master Bedroom

- **Negative Side (Damage):** Skirting-level dampness observed along the base of walls.

Source: Sample Report | Page 3

Source: Sample Report | Page 3

Source: Sample Report | Page 3

- **Positive Side (Source):** Master Bedroom Bathroom — tile hollowness detected, gaps in tile joints, gaps around Nahani trap, and loose plumbing joints with visible rust around edges.

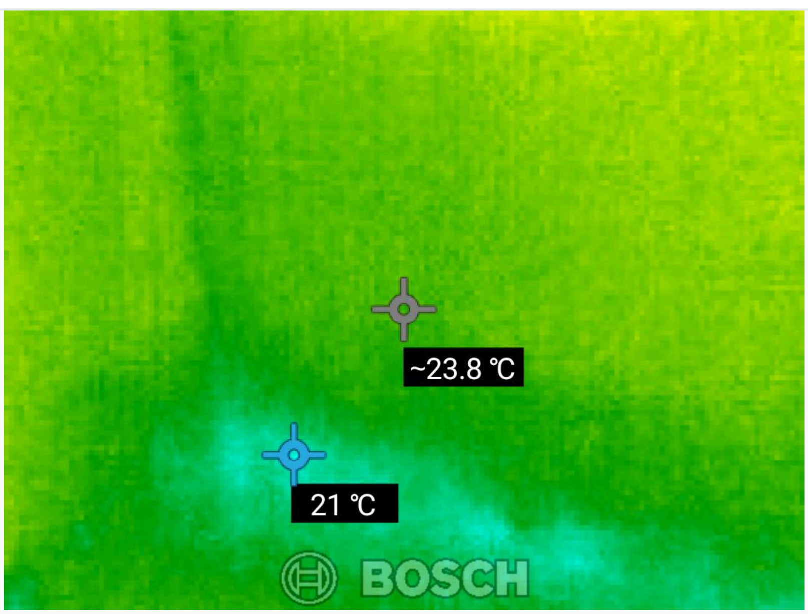
Source: Sample Report | Page 3

UrbanRoof

Source: Sample Report | Page 4

Source: Sample Report | Page 4

- **Leakage Details:** Continuous ("All time") leakage confirmed. Cause attributed to concealed plumbing defects and damaged Nahani trap/brickbat coba layer beneath the tile flooring.
- **Thermal Findings:** Thermal scans show hotspot of 25.7°C and coldspot of 20.7°C, confirming active moisture zones.



Source: Thermal Images | Page 1

Impacted Area 4: Kitchen

- **Negative Side (Damage):** Skirting-level dampness observed.

Source: Sample Report | Page 3

Source: Sample Report | Page 3

- **Positive Side (Source):** Traced to the Master Bedroom Bathroom, which shares a common wall with the kitchen. Water from the bathroom is migrating laterally through the wall structure.

Source: Sample Report | Page 3

Source: Sample Report | Page 3

- **Thermal Findings:** Temperature range of 20.8°C to 25.8°C recorded, supporting the presence of moisture within the shared wall.

Source: Thermal Images | Page 1

Impacted Area 5: Master Bedroom — Wall Dampness

- **Negative Side (Damage):** Wall dampness observed on the external-facing wall of the Master Bedroom, separate from the skirting-level issue.

Source: Sample Report | Page 3

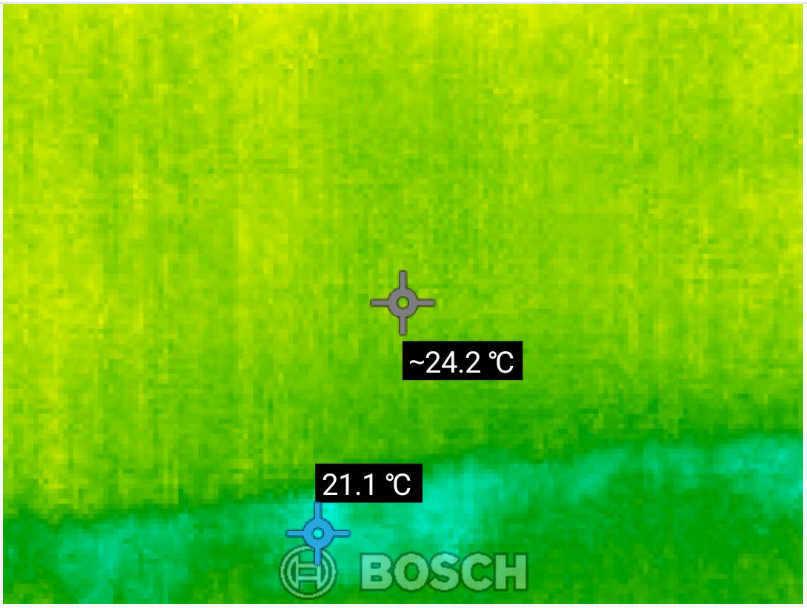
Source: Sample Report | Page 3

- **Positive Side (Source):** External wall cracks and duct issues identified. Moderate major and minor cracks were observed on the external surface, along with cracked and leaking external plumbing pipes. Algae, fungus, and moss growth on the external wall confirms prolonged moisture exposure.

Source: Sample Report | Page 3

Source: Sample Report | Page 4

- **Thermal Findings:** Thermal imaging shows temperature differential of approximately 5°C (hotspot 26.5°C, coldspot 21.5°C) along the affected wall, indicating water ingress through external cracks.



Source: Thermal Images | Page 1

Impacted Area 6: Parking Area

- **Negative Side (Damage):** Active seepage observed in the parking area, likely dripping from the ceiling/slab above.

Source: Sample Report | Page 3

- **Positive Side (Source):** Linked to the Common Bathroom above — tile hollowness and plumbing issues in the bathroom are allowing water to percolate through the floor slab into the parking area below.

Source: Sample Report | Page 3

- **Thermal Findings:** Temperature range of 21.2°C to 26.2°C recorded in surrounding areas, consistent with moisture migration through the slab.

Source: Thermal Images | Page 1

Impacted Area 7: Common Bathroom — Ceiling Dampness

- **Negative Side (Damage):** Ceiling dampness observed in the Common Bathroom. Additionally, outlet leakage was noted.

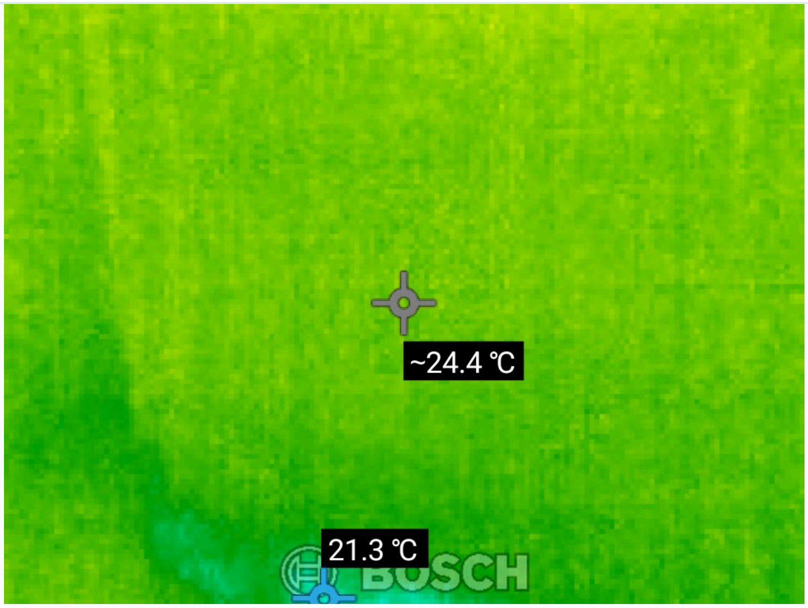
Source: Sample Report | Page 3

- **Positive Side (Source):** The leakage originates from Flat No. 203 (the unit directly above). Open tile joints and outlet leakage in Flat 203's bathroom are causing water to seep through the floor slab into this unit's ceiling.

Source: Sample Report | Page 3

Source: Sample Report | Page 3

- **Thermal Findings:** Temperature readings of hotspot 26.0°C and coldspot 21.0°C recorded, confirming active moisture in the ceiling slab.



Source: Thermal Images | Page 1

External Wall — General Observations

- Moderate major and minor cracks observed on the external building surface
- External plumbing pipes are moderately cracked and leaking
- Moderate algae, fungus, and moss growth observed on external walls, confirming chronic moisture exposure
- Existing type of paint and manufacturer: Not sure
- Structural Condition of RCC Members: 100%
- Condition of cracks on RCC Column and Beam: Moderate
- Condition of rust marks on RCC Beam and Column: N/A

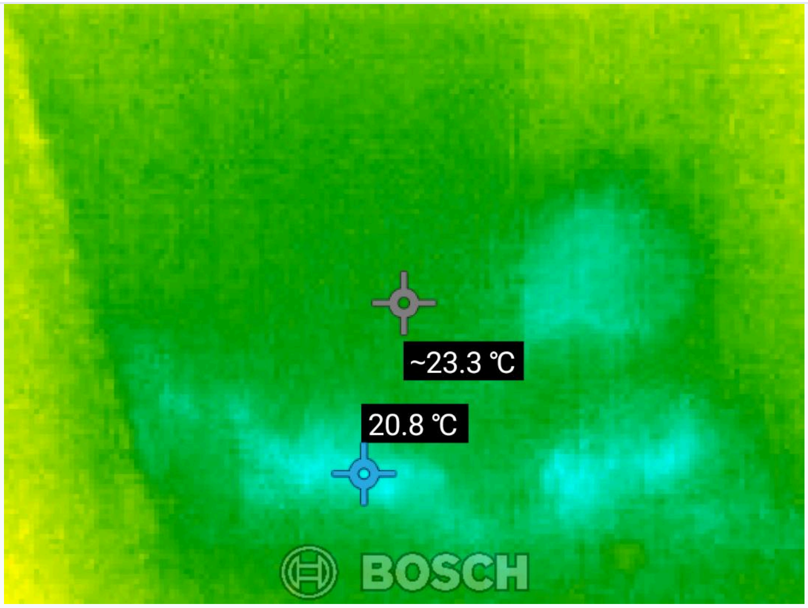
Thermal Imaging Summary

All thermal scans were performed on 27/09/2022 using a **Bosch GTC 400 C Professional** thermal camera (Serial: 02700034772) with an emissivity setting of 0.94 and reflected temperature of 23°C.

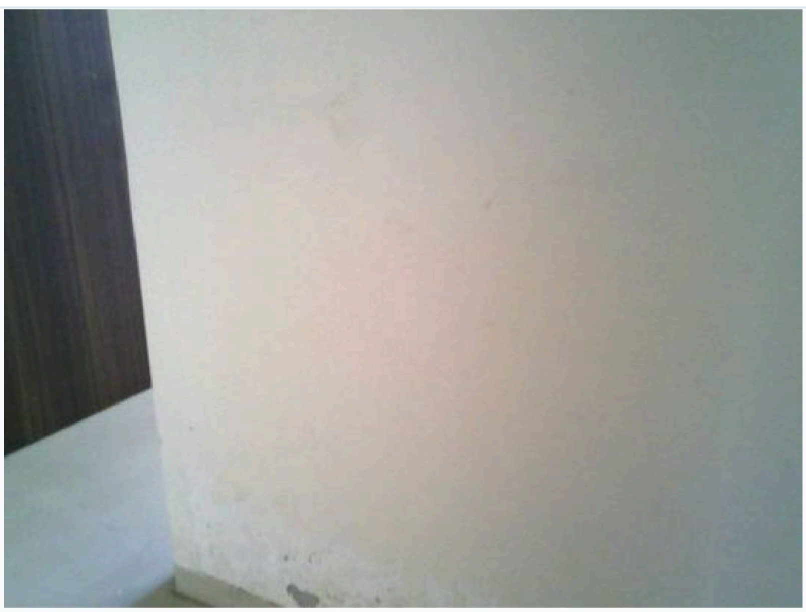
| Scan | Hotspot | Coldspot | Differential | |------|---------|----------|-------------| | RB02380X | 28.8°C | 23.4°C | 5.4°C | | RB02386X | 27.4°C | 22.4°C | 5.0°C | | RB02395X | 27.0°C | 22.0°C | 5.0°C | | RB02402X | 25.7°C | 20.7°C | 5.0°C | | RB02403X | 25.5°C | 20.5°C | 5.0°C | | RB02404X | 25.8°C | 20.8°C | 5.0°C | | RB02405X | 26.2°C | 21.2°C | 5.0°C | | RB02406X | 26.5°C | 21.5°C | 5.0°C | | RB02392X | 25.8°C | 20.8°C | 5.0°C | | RB02393X | 25.9°C | 20.9°C | 5.0°C | | RB02394X | 26.0°C | 21.0°C | 5.0°C | | RB02396X | 26.1°C | 21.1°C | 5.0°C | | RB02397X | 26.3°C | 21.3°C | 5.0°C | | RB02398X | 25.6°C | 20.6°C | 5.0°C | | RB02399X | 26.5°C |

21.5°C | 5.0°C | | RB02400X | 25.2°C | 20.2°C | 5.0°C | | RB02401X | 25.6°C | 20.6°C | 5.0°C | | RB02381X | 27.0°C | 22.0°C | 5.0°C |

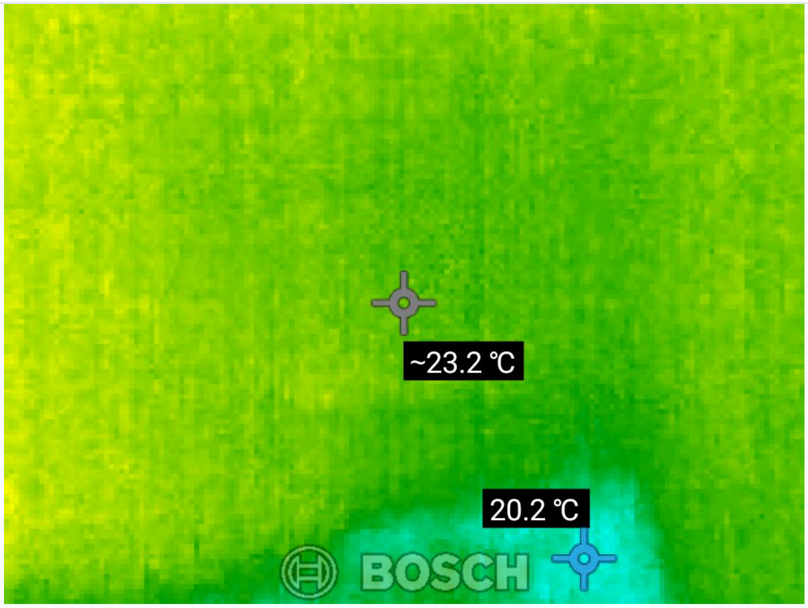
Temperature differentials of 5.0–5.4°C consistently indicate subsurface moisture presence across all scanned zones.



Source: Thermal Images | Page 1



Source: Thermal Images | Page 1



Source: Thermal Images | Page 1

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3. PROBABLE ROOT CAUSE

Cause 1: Internal Bathroom Waterproofing Failure

The bathrooms (Common and Master Bedroom) show tile hollowness, indicating the adhesive bond between tiles and the substrate has failed. This creates voids where water accumulates and migrates laterally through walls to adjacent rooms (Hall, Bedroom, Kitchen). The damaged Nahani trap and brickbat coba layer beneath the flooring allows water to percolate vertically into lower floors (Parking Area).

Cause 2: Concealed Plumbing Defects

Loose plumbing joints with visible rust around fixtures (flush tank, shower, angle cock, bibcock, washbasin) indicate aging and degraded pipe connections. These joints leak continuously ("All time"), feeding moisture into the surrounding masonry.

Cause 3: External Envelope Breaches

Moderate cracks on the external wall allow rainwater ingress into the Master Bedroom walls. Cracked external plumbing pipes add to the problem. The presence of algae, fungus, and moss confirms the wall has been absorbing moisture for an extended period.

Cause 4: Inter-floor Leakage from Flat No. 203

Open tile joints and outlet leakage in the flat above (203) are causing water to seep through the floor slab, resulting in ceiling dampness in the Common Bathroom of the subject property.

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4. SEVERITY ASSESSMENT

| Issue | Severity | Reasoning | |-------|----------|-----------| | Bathroom waterproofing failure | **Critical** | Active continuous leakage affecting 4+ areas. Progressive damage will worsen without intervention. | | Concealed plumbing leaks | **Critical** | "All time" leakage indicates constant water flow. Risk of structural deterioration and mold growth. | | External wall cracks | **High** | Allowing rainwater ingress. Algae/moss growth confirms chronic exposure. Will worsen with each monsoon season. | | External pipe leakage | **High** | Contributing to wall dampness and biological growth on external surface. | | Inter-floor leakage (Flat 203) | **High** | Active ceiling dampness. Requires coordination with neighboring flat for resolution. | | Parking area seepage | **Medium** | Downstream effect of bathroom leakage. Will resolve once primary source is fixed. |

Overall Property Severity: HIGH

Multiple critical and high-severity issues with active, continuous water ingress require immediate remediation to prevent structural damage, mold growth, and health hazards.

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5. RECOMMENDED ACTIONS

Priority 1 — Immediate (Within 2 weeks)

1. **Bathroom Waterproofing Overhaul (Common & Master Bedroom Bathrooms)** - Remove all existing tiles in both bathrooms - Inspect and repair/replace the Nahani trap and brickbat coba layer - Apply fresh waterproofing membrane (polymer-modified cementitious coating recommended) - Re-tile with proper adhesive and sealed grout joints - Conduct water retention test (48-hour ponding test) before finishing

2. **Plumbing Fixture Replacement** - Replace all corroded/loose plumbing joints (flush tank, shower, angle cock, bibcock, washbasin connections) - Pressure-test concealed plumbing lines to identify hidden leaks - Replace any damaged concealed pipes

Priority 2 — Short-term (Within 1 month)

3. **External Wall Repair** - Fill all external wall cracks using appropriate crack-filling compound - Repair/replace cracked external plumbing pipes - Clean and treat biological growth (algae, fungus, moss) with anti-fungal solution - Apply waterproof external coating after repairs

4. **Inter-floor Coordination (Flat 203)** - Notify building management about leakage from Flat 203 - Request inspection and repair of open tile joints and outlet in Flat 203's bathroom - Monitor ceiling dampness after repairs are completed upstairs

Priority 3 — After Source Remediation

5. **Internal Restoration** - Allow all damp walls to dry completely (minimum 2-3 weeks after source repair) - Remove damaged plaster and skirting from affected areas - Apply anti-dampness primer before re-plastering - Re-plaster and repaint all affected internal surfaces

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6. ADDITIONAL NOTES

- The inspection score of 85.71% with 1 flagged item (WC and External Wall categories) indicates that while the property's general structural condition is acceptable, the water-related issues are significant enough to require urgent attention.
- All thermal scans were performed with consistent equipment settings (Emissivity: 0.94, Reflected Temperature: 23°C), ensuring reliable and comparable readings across all zones.
- The consistent 5.0°C temperature differential across nearly all scans suggests a uniform moisture distribution pattern, indicating the leakage has been active for a significant period.
- No previous structural audit or repair work has been performed on this property, meaning these issues have likely been developing unchecked.
- The thermal imaging device used (Bosch GTC 400 C Professional) provides adequate resolution for building diagnostic purposes.

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7. MISSING OR UNCLEAR INFORMATION

- **Customer Name:** Not provided in the inspection report
- **Customer Contact (Mobile/Email):** Not provided
- **Customer Address:** Not provided
- **Property Age:** Not specified (field left blank)
- **Bedroom Negative-Side Photos:** No specific negative-side photographs were listed for Impacted Area 2 (Bedroom skirting level dampness), though positive-side photos were referenced as Photo 15-19
- **Thermal Image Location Mapping:** The thermal report does not specify which physical area each thermal scan corresponds to, making precise area-to-scan mapping not possible
- **Conflicting Structural Data:** The report states "Structural Condition of RCC Members: 100%" but also notes "Condition of cracks on RCC Column and Beam: Moderate." These two findings appear

contradictory — moderate cracking on RCC elements would typically not correspond to a 100% structural score. This requires clarification from the inspection team.

- **Corrosion/Spalling Data:** The field for "Condition of corrosion/spalling of concrete/exposed" appears to be cut off in the source document, and the data is incomplete.

This report was generated using an AI-powered diagnostic system.

All findings are based strictly on the provided Inspection Report and Thermal Report.

No information has been fabricated. Missing data is explicitly marked as "Not Available".